22 There are only green pens and blue pens in a box.

There are three more blue pens than green pens in the box.
There are more than 12 pens in the box.

Simon is going to take at random two pens from the box.

The probability that Simon will take two pens of the same colour is $\frac{27}{55}$

Work out the number of green pens in the box.

Let green = x, blue = $x + 3$, total = $2x + 3$.

$P(G) = \dfrac{x}{2x + 3}$

1 mark
Express a probability using x

$P(GG) = \dfrac{x}{2x + 3} \times \dfrac{x - 1}{2x + 2}$

1 mark
Without replacement: reduce both counts

$P(BB) = \dfrac{x + 3}{2x + 3} \times \dfrac{x + 2}{2x + 2}$

$$\frac{x(x - 1) + (x + 3)(x + 2)}{(2x + 3)(2x + 2)} = \frac{27}{55}$$

1 mark
Add the two same-colour cases

	x	2
x	x^2	$2x$
3	$3x$	6

	$2x$	2
$2x$	$4x^2$	$4x$
3	$6x$	6

$55(2x^2 + 4x + 6) = 27(4x^2 + 10x + 6)$

$110x^2 + 220x + 330 = 108x^2 + 270x + 162$

$2x^2 - 50x + 168 = 0$, so $x^2 - 25x + 84 = 0$

Product 84, sum −25: −21 and −4.

1 mark
Clear denominators and simplify

	x	−21
x	x^2	$-21x$
−4	$-4x$	84

$(x - 21)(x - 4) = 0$

1 mark
Solve the quadratic

$x = 4$ gives only 11 pens; reject.

$x = 21$ gives 45 pens, so 21 green pens.

1 mark
Use the condition: more than 12 pens

21

(Total for Question 22 is 6 marks)

TOTAL FOR PAPER IS 80 MARKS

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